Binomial Distribution

Section 1: Binomial Criteria & Single Probabilities

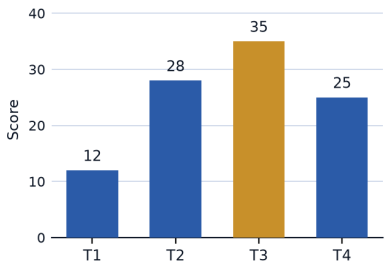
1 Which of the following is NOT a required condition for a variable X to have a binomial distribution $B(n, p)$?

a. The probability of success changes with each trial

b. There is a fixed number of trials n

c. Each trial results in one of only two outcomes (success or failure)

d. Trials are independent of one another



2 If $X \sim B(6, 0.3)$, calculate $P(X = 3)$ correct to 4 decimal places.

a. 0.1852

b. 0.2541

c. 0.3087

d. 0.1176

3 If $X \sim B(5, 0.4)$, find $P(X = 0)$.

a. 0.0778

b. 0.0102

c. 0.0625

d. 0.1000

Section 2: Cumulative Binomial Probabilities

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4 If $X \sim B(4, 0.5)$, find $P(X \geq 2)$.

a. $\frac{11}{16} = 0.6875$

b. $\frac{1}{2} = 0.5000$

c. $\frac{5}{16} = 0.3125$

d. $\frac{3}{8} = 0.3750$

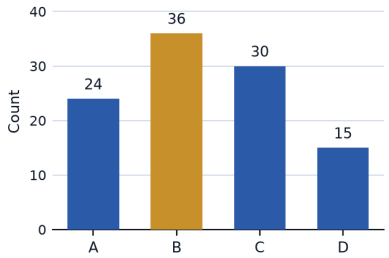
5 A manufacturer knows that 5% of lightbulbs are defective. In a random sample of 20 bulbs, what is the probability that at most 1 is defective?

a. 0.7358

b. 0.6824

c. 0.8245

d. 0.9128



6 A fair coin is tossed 6 times. Find the probability of obtaining at least 5 heads.

a. $\frac{7}{64} \approx 0.1094$

b. $\frac{6}{64} \approx 0.0938$

c. $\frac{1}{64} \approx 0.0156$

d. $\frac{11}{64} \approx 0.1719$

Section 3: Mean & Variance of the Binomial Distribution

7 A random variable $X \sim B(48, 0.25)$. Find its mean and standard deviation.

a. Mean $= 12, \sigma = 3$

b. Mean $= 12, \sigma = 9$

c. Mean $= 16, \sigma = 4$

d. Mean $= 12, \sigma = \sqrt{3}$

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8 For a binomial distribution $X \sim B(n, p)$, the mean is 6 and the variance is 3.6. Find n and p.

a. $n = 15, p = 0.4$

b. $n = 20, p = 0.3$

c. $n = 12, p = 0.5$

d. $n = 24, p = 0.25$

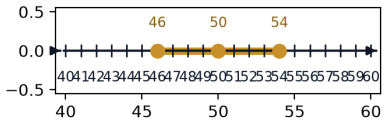
9 What is the maximum possible variance of a binomial distribution with $n = 40$ trials?

a. 10

b. 20

c. 5

d. 40



Section 4: Contextual Modeling & Problem Solving

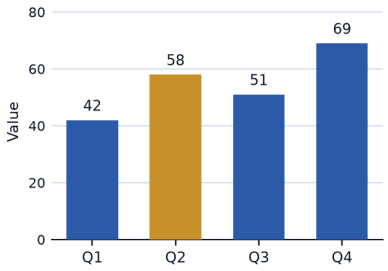
10 A treatment is successful for 80% of patients. If 10 patients are treated, what is the most likely number of successful outcomes (the mode)?

a. 8

b. 7

c. 9

d. 10



11 How many times must a fair die be rolled so that the probability of getting at least one 6 is greater than 0.9 ?

a. At least 13 rolls

b. At least 10 rolls

c. At least 6 rolls

d. At least 18 rolls

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12 In an exam, 70% of candidates pass. Out of 8 randomly chosen candidates, what is the probability that exactly 6 pass, given that at least 5 pass?

a. ≈ 0.380

b. ≈ 0.520

c. ≈ 0.296

d. ≈ 0.420

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